

# Aurora X200 Installation and Maintenance Guide

*Document AX200-IM-003, Revision C, April 2026. For certified Halcyon technicians.*

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## 1. About This Guide

This guide describes how to install, commission and maintain an Aurora X200 autonomous inspection system consisting of the AX200-BASE-01 aircraft and the AX200-DOCK-2 docking station. It applies to aircraft running AuroraOS 4.2.x and docks running DockOS 2.6.x. Installation must be performed by a technician who has completed the Halcyon Certified Installer course (course code HCI-200).

Warnings in this guide use three levels: DANGER indicates a risk of serious injury, WARNING indicates a risk of equipment damage and NOTICE indicates important operational information. Always remove both battery packs before working on propellers or motors. Specifications referenced here are taken from the Aurora X200 Technical Specification, document AX200-DS-004.

## 2. Unboxing Checklist

Inspect the shipping crate and the AX200-CASE-HD hard case for damage before signing the delivery note. Transport damage must be reported to Halcyon Support within 48 hours of delivery with photographs. A standard system shipment contains the following items:

- ? 1 Aurora X200 aircraft (AX200-BASE-01) with the standard sensor gimbal and gimbal transport lock installed.
- ? 2 battery packs AX200-BAT-12K, shipped at 50 percent charge.
- ? 1 docking station AX200-DOCK-2 on a wooden pallet, with lid clamps fitted.
- ? 2 sets of propellers AX200-PROP-CF (one installed, one spare).
- ? 1 RTK base antenna with 5 m cable and mast bracket.
- ? 1 installer tool kit: 3 mm and 5 mm hex keys, 2 to 10 Nm torque wrench, propeller gauge and lens cloth.
- ? 1 mounting template and 4 M12 x 120 mm stainless steel anchor bolts with washers.
- ? 2 SIM adapters, 1 USB-C console cable and the quick start card.

Record the aircraft serial number (label inside the battery bay) and the dock serial number (label behind the service panel) in the commissioning report. Both numbers are needed to register the system in Halcyon Insight and to make warranty claims.

### 3. Site Requirements for the Docking Station

The dock must be installed on a level reinforced-concrete pad at least 1,600 x 1,400 mm and 150 mm thick, or on a Halcyon-approved steel frame. The pad must be level to within 2 degrees in both axes; the dock feet allow a further 15 mm of fine adjustment. Keep a clear radius of 20 m around the dock free of trees, masts and overhead lines to allow safe vertical take-off and landing.

Power requirements are a dedicated 240 V AC, 16 A single-phase circuit with a residual-current device, terminated in the weatherproof inlet on the rear of the dock. Network connectivity is provided by Gigabit Ethernet or the built-in 5G modem; a minimum sustained uplink of 20 Mbit/s is recommended for same-day data upload. The RTK base antenna must have an unobstructed sky view and be mounted within 5 m of the dock.

### 4. Mounting the Docking Station

Place the mounting template on the pad, orient the lid hinge away from the prevailing wind and mark the four bolt positions. Drill 14 mm holes to a depth of 130 mm, clean them with compressed air and insert the four M12 x 120 mm stainless steel anchor bolts using the supplied chemical anchor capsules. Allow the anchors to cure for the time stated on the capsule (typically 45 minutes at 20 °C).

Lift the dock onto the bolts using a forklift or four people (the dock weighs 96 kg) and fit the washers and nuts. Tighten the nuts in a cross pattern to a torque of 65 Nm using a calibrated torque wrench. Connect the grounding strap to the site earth point, remove the lid clamps and check that the lid opens fully without obstruction. Re-check bolt torque after the first 7 days of operation and then at every 6-month service.

### 5. Initial Power-Up and Network Configuration

Insert the SIM cards, connect the Ethernet cable if used and switch on the dock. Boot takes about 90 seconds; the status LED shows solid amber while booting, blinking blue when waiting for configuration and solid green when online. Connect a laptop to the USB-C console port and open the DockOS setup page at the address printed on the quick start card.

In the setup wizard, set the time zone, the site name and the network mode, then sign in with the Halcyon Insight installer account to claim the dock. Place the aircraft on the landing pad with both batteries installed and press Pair. Pairing takes approximately 2 minutes. If the aircraft firmware is older than AuroraOS 4.2.1, the dock downloads and installs the update automatically; do not power off during the update.

## 6. Calibration Procedure

Full calibration takes about 25 minutes and is required at commissioning, after any propeller or motor replacement, after every firmware update and whenever error code E-310 (camera calibration drift) is reported. Perform the following steps in order from the Calibration menu in Halcyon Insight:

? Step 1, IMU calibration: place the aircraft on a level surface and hold it in each of the 6 requested orientations for 10 seconds.

? Step 2, Compass calibration: at least 10 m from steel structures, rotate the aircraft slowly through 3 complete figure-eight patterns until the app confirms success.

? Step 3, LiDAR-camera boresight: place the supplied calibration target board exactly 5 m in front of the gimbal and capture 12 frames as prompted.

? Step 4, RTK survey-in: with the base antenna connected, run a 15-minute survey-in and confirm a horizontal accuracy below 2 cm.

? Step 5, Dock alignment: command an automatic landing and verify the aircraft settles on the charging contacts within 5 mm of centre.

Calibration results are stored in the aircraft and uploaded to Halcyon Insight. A calibration flight of at least 5 minutes should follow, during which vibration and positioning quality are checked automatically.

## 7. Maintenance Schedule

Maintenance intervals are expressed in flight hours or calendar time, whichever occurs first. Halcyon Insight tracks flight hours per component and raises a maintenance task automatically. The intervals are:

Pre-flight visual inspection: Automatic before every flight.

Propeller inspection: Every 25 flight hours.

Propeller replacement: Every 150 flight hours or 12 months.

Motor bearing inspection: Every 300 flight hours.

Battery health check: Every 50 charge cycles.

Battery replacement: After 400 cycles or when capacity falls below 80 percent.

Dock air filter replacement: Every 6 months.

Dock bolt torque check: After 7 days, then every 6 months.

Full certified service: Every 600 flight hours or 24 months.

Firmware update: Quarterly, or when a security release is published.

The full certified service must be performed by a Halcyon-certified technician and includes motor replacement if bearing play exceeds 0.2 mm, gimbal inspection, seal replacement on the dock lid and a complete calibration.

## 8. Propeller Replacement

Replace all four propellers as a set using part number AX200-PROP-CF. Power off the aircraft and remove both battery packs. Clockwise propellers are marked with a silver ring and counter-clockwise propellers with a black ring; fitting a propeller on the wrong motor will trigger error E-455 on the next start-up.

Remove the old propeller with the 5 mm hex key, clean the motor hub, fit the new propeller with a new locking washer and tighten the bolt to 4.5 Nm. After fitting all four, run the propeller balance test from the Maintenance menu and then complete the calibration procedure in section 6. Dispose of damaged propellers as carbon-fibre waste; they must not be repaired.

## 9. Battery Care

AX200-BAT-12K packs charge in the dock from empty to 100 percent in 55 minutes. Charging is inhibited when the pack temperature is below -10 °C or above +40 °C; the dock heats or cools the battery bay first, which can add up to 20 minutes in extreme weather. Packs stored outside the dock for more than 7 days should be kept at 50 to 60 percent charge in a dry place between 10 °C and 25 °C.

Halcyon Insight records the cycle count and measured capacity of every pack. A pack should be retired after 400 cycles or when its capacity falls below 80 percent of nominal. Swollen, punctured or dropped packs must be taken out of service immediately and returned in the supplied fire-resistant bag.

## 10. Commissioning Flight and Handover

After calibration, perform a 10-minute commissioning flight using the built-in Commissioning mission, which climbs to 30 m, flies a 50 m square, captures a test image set and lands on the dock. Verify in Halcyon Insight that positioning accuracy stayed below 5 cm, that vibration remained in the green band and that all three sensors produced valid data.

Complete the commissioning report, including both serial numbers, the bolt torque value, the calibration results and photographs of the installed dock, and submit it through the installer portal within 5 business days. Warranty coverage starts on the delivery date, but claims are only accepted once the commissioning report is on file. Hand the quick start card and the customer administrator credentials to the customer in a sealed envelope.

## 11. Cleaning, Storage and Transport

Clean the LiDAR window, camera lens and thermal window with the supplied lens cloth and isopropyl alcohol at every 25-hour inspection or whenever error E-302 appears. Never use compressed air on the gimbal. Wipe the dock charging contacts with a dry cloth and check the lid seal for cracks every 6 months.

For storage longer than 30 days, remove the batteries, bring them to 50 to 60 percent charge and store the folded aircraft in the AX200-CASE-HD case between -30 °C and +60 °C. For transport by air, battery packs must be at 30 percent state of charge or lower and shipped as UN 3481 dangerous goods with the UN 38.3 test summary, which is available on the Halcyon Support portal.

## 12. Error Codes

Error codes are shown in Halcyon Insight and in the dock status page. The most common codes and their meanings are:

E-101: Battery charge below 15 percent; aircraft returns to dock immediately.

E-114: Battery temperature out of range; charging or take-off inhibited.

E-207: GNSS lock lost; aircraft holds position using visual odometry.

E-215: RTK correction timeout; positioning accuracy degraded to 1.5 m.

E-302: LiDAR return degraded; clean the LiDAR window and re-run boresight.

E-310: Camera calibration drift; run the full calibration procedure.

E-401: Motor overcurrent; inspect the affected motor and propeller.

E-455: Propeller imbalance or wrong rotation direction detected.

E-503: Dock charger fault; check the 240 V supply and charger fuse F2.

E-520: Dock lid obstruction; clear debris and test lid travel.

E-601: Command link lost for more than 10 seconds; return to dock.

E-999: Firmware integrity check failed; unsigned image refused.

Codes E-401, E-503 and E-999 require a support ticket before the system can be returned to service. All other codes clear automatically once the underlying condition is resolved and a successful self-test has completed.

## 13. Warranty Terms

Halcyon Dynamics warrants the Aurora X200 aircraft and the AX200-DOCK-2 docking station against defects in materials and workmanship for 24 months from the date of delivery. Battery packs are warranted for 12 months or 300 charge cycles, whichever occurs first. Customers who purchase the AuroraCare Plus plan within 90 days of delivery extend aircraft and dock coverage to 36 months and receive one preventive service visit per year.

The warranty does not cover:

- ? Crash damage resulting from manual piloting error or disabled geofencing.
- ? Operation outside the specified temperature range of -20 °C to +45 °C or wind limits of 14 m/s.
- ? Use of unsigned or modified firmware, or propellers other than AX200-PROP-CF.
- ? Saltwater immersion, lightning strike or other external events.
- ? Cosmetic wear, and consumables such as propellers, filters and seals.
- ? Installations not performed by a Halcyon Certified Installer.

To make a claim, open a ticket in the Halcyon Support portal within 30 days of discovering the defect, quoting the aircraft or dock serial number and attaching the relevant flight log. Approved returns receive a return material authorisation (RMA) number and are repaired or replaced within 10 business days of receipt at the Meridian Falls service centre.

## **14. Technical Support**

Halcyon Technical Support is available Monday to Friday from 06:00 to 20:00 Mountain Time and on Saturday from 08:00 to 16:00 Mountain Time. Support is closed on Sundays and on United States federal holidays. Contact support by telephone at +1 555 0100 220, by email at [support@halcyon-dynamics.example](mailto:support@halcyon-dynamics.example) or through the Halcyon Support portal.

Response targets depend on ticket priority: Priority 1 (system down) receives a response within 4 business hours, Priority 2 (degraded operation) within 1 business day and Priority 3 (questions and feature requests) within 3 business days. AuroraCare Plus customers receive 24-hour telephone coverage for Priority 1 issues and a dedicated support engineer.